24 FEB 2026

# AADK TASK 1

Composables Practice Reflection & Planning

MindMatrix

## Problem 1: Greeting Card Layout (Using Column + Text)

### 1. Problem Summary

This exercise asked to build a simple greeting screen using multiple Text composables arranged vertically. The screen included a title message and a short description below it. The layout had to be centered properly on the screen.

### 2. My Planned Approach

Before writing code, I planned the structure like this:

- Use Column as the main layout container.
- Add two Text composables inside it:
  - One for the title
  - One for the body text
- Apply:
  - Modifier.fillMaxSize() to use full screen
  - verticalArrangement = Arrangement.Center
  - horizontalAlignment = Alignment.CenterHorizontally
  - Padding for spacing
- Increase font size for title and use bold style.

### 3. Actual Result & Reflection

After implementing it:

- The Column worked well for vertical arrangement.
- Centering both texts required proper alignment settings.
- Spacing between texts was slightly tricky at first.
- I initially forgot padding, so the UI looked cramped.

What I learned:
Planning the layout structure first made coding easier. Next time, I will decide spacing and font sizes more clearly before starting.

## Problem 2: Profile Row Layout (Using Row + Image + Text)

### 1. Problem Summary

This exercise required creating a horizontal layout with an image and text placed side by side. The image represented a profile picture, and the text displayed user information.

### 2. My Planned Approach

Before coding, I planned:

- Use Row as the main container.
- Add:
  - Image composable for profile picture.
  - Column inside the Row for name and description.
- Apply:
  - Modifier.padding(16.dp)
  - verticalAlignment = Alignment.CenterVertically
  - Fixed size for image (e.g., 64.dp)
- Use spacing between image and text.

### 3. Actual Result & Reflection

After implementing:

- Row helped arrange items horizontally as expected.
- Managing spacing between image and text needed Spacer.
- I learned that image size strongly affects alignment.
- Adding contentDescription for accessibility was important.

What I would improve next time:
I would sketch the layout first and decide image size clearly before coding.

## Planning Checklist Template for New Composable UI Problems

When solving a new Compose UI problem, I will follow this checklist:

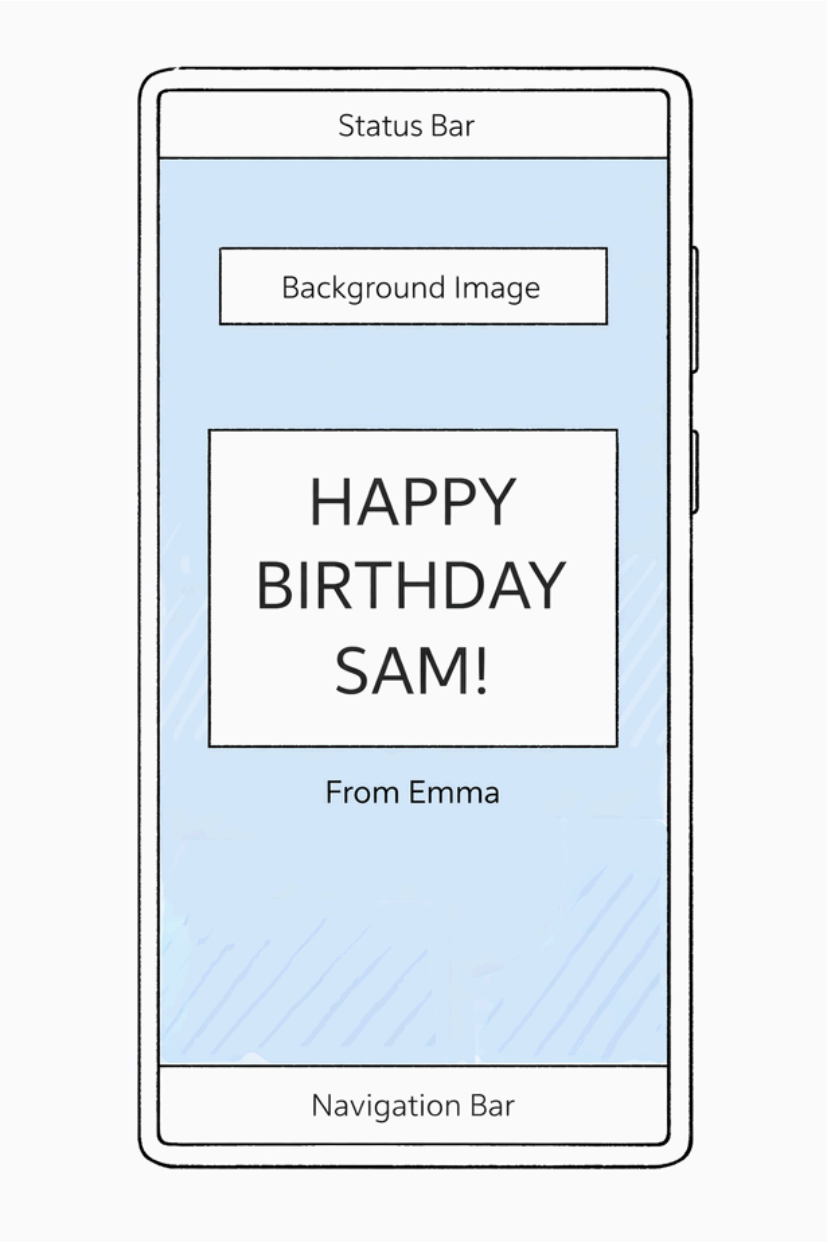
**Step 1: Understand the Specification**

- What UI elements are required?
- Is the layout vertical, horizontal, or mixed?

**Step 2: Sketch a Wireframe**

- Draw rough layout on paper.

Identify sections (header, body, footer).



**Step 3: Choose Composables**

- Column for vertical layout
- Row for horizontal layout
- Box for overlapping elements
- Text, Image, Button as needed

**Step 4: Plan Modifiers**

- Padding and spacing (dp values)
- Alignment settings
- Size constraints
- fillMaxWidth / fillMaxSize

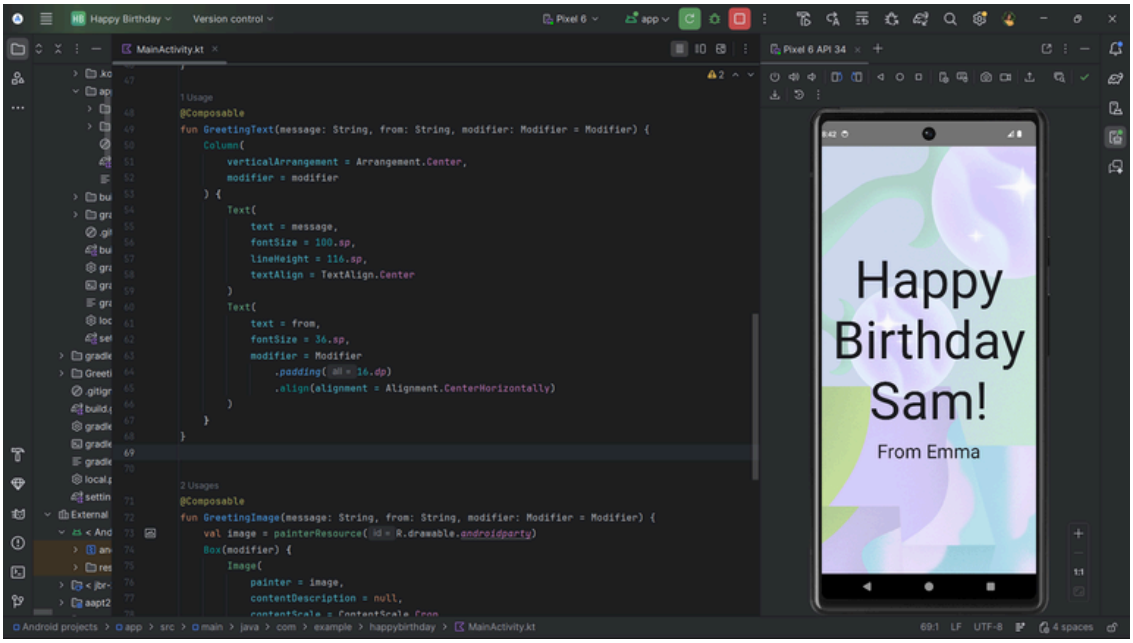
**Step 5: Consider Accessibility**

- Add contentDescription for images
- Ensure text is readable (minimum 16sp)
- Maintain proper color contrast
- Button minimum height 48dp

**Step 6: Test Responsiveness**

- Test on small and large screens
- Check if layout looks crowded
- Adjust spacing if needed

## Short Reflection

Planning before coding helps me clearly understand layout structure and reduces confusion. It makes me more confident while writing Compose code because I already know which composables to use. Reflection after implementation helps me identify mistakes and improve future designs. Overall, planning and reflection make my UI development faster, cleaner, and more professional.

## Screenshot